



Carisurg

Assignment 4

Report

Focus Topic: MAPS — Mean Arterial Pressure.

Due Date: May 24th, 2025.

Prepared by **Josiah-John Green**

Mean Arterial Pressure (MAP)

Owing to my introduction to MAPS in the previous activity, I felt it was only prudent to write a report on it, explaining not only the clinical value but also its range and significance in triaging.

That being said, **Mean Arterial Pressure (MAP)** is described as a calculated measure of the average pressure driving blood through the arteries across a full cardiac cycle.

It is derived from **systolic** and **diastolic blood pressure** using the formula:

$$\text{MAP} = (\text{SBP} + 2 \times \text{DBP}) / 3,$$

and is expressed in **millimetres of mercury (mmHg)**.

The normal clinical range for MAP in a **healthy adult** is **70–100 mmHg**.

A MAP **below 65 mmHg** is the key threshold for **organ hypoperfusion** — the point at which vital organs such as the kidneys, brain, and heart may no longer be receiving adequate blood flow. This threshold is a central criterion in the diagnosis and management of septic shock—an extreme immune reaction, causing blood vessels to dilate and leak, resulting in organ failure and tissue damage.

However, a MAP consistently above 110 mmHg may **indicate hypertensive crisis**, particularly when accompanied by end-organ symptoms, resulting in strokes and heart attacks.

In an emergency department triage setting, **MAP is one of the most clinically informative single values** a medical professional can calculate. Unlike SBP or DBP in isolation, MAP accounts for the **full pressure cycle** and gives a **truer picture of perfusion status**.

For **example**, a patient arriving with an **SBP of 85 mmHg** and a **DBP of 50 mmHg** has a MAP of approximately **62 mmHg** — firmly below the MAP threshold — and should be escalated immediately regardless of how alert they appear.

This makes MAP especially valuable for identifying patients in compensated shock, where the body is maintaining consciousness by redirecting blood flow, masking the severity of their condition.

At Mercer General, where MAP is not always directly measured at triage, calculating it from the recorded SBP and DBP values is a simple but powerful step toward earlier recognition of deteriorating patients.